

E-commerce Data Analytics — Business Report

SQL | Python | Power BI

1. Executive Summary

This project analyses a large-scale e-commerce dataset containing approximately 1 million orders across customers, products, sellers, payments and customer reviews.

The analysis combines SQL, Python and Power BI to evaluate business performance, product profitability, customer behaviour, payment patterns and operational performance.

The dataset contains 1,000,000 orders and generates approximately \$983.14 million in product sales revenue and \$251.50 million in gross profit, resulting in an overall gross margin of approximately 25.58%.

- Revenue growth in recent years has been primarily driven by increasing order volume, while average order value remained relatively stable.
- Electronics is the dominant revenue category, contributing approximately 74.5% of total product sales revenue, but its gross margin is comparatively low at approximately 18.2%.
- Repeat customers represent approximately 84.1% of customers but contribute approximately 95.5% of revenue.
- Product revenue is concentrated among a relatively small proportion of products, with the top 10% generating approximately 66.7% of total revenue.
- Delivery performance is associated with customer satisfaction: lower review scores are observed alongside longer delivery times and higher late-delivery rates.
- Credit cards are the dominant payment method, representing approximately 50.7% of total payment value.

Overall, the analysis suggests that the business has strong revenue scale and a substantial repeat-customer base, while profitability management, product concentration and delivery performance represent important areas for further investigation.

2. Business Performance

2.1 Overall Performance

The business generated approximately \$983.14 million in product sales revenue across 1 million orders. Average order value was approximately \$983.14, while gross profit reached approximately \$251.50 million, producing an overall gross margin of approximately 25.58%. These results indicate that the business operates at significant scale, with a relatively stable value generated per order.

2.2 Revenue Trend

Revenue experienced a significant decline between 2019 and 2021, followed by a strong recovery from 2022 onwards.

Year	Revenue	Orders	AOV
2019	\$112.42M	113,558	\$989.98
2020	\$44.68M	45,553	\$980.89

2021	\$44.34M	45,501	\$974.43
2022	\$111.32M	113,886	\$977.43
2023	\$168.35M	170,700	\$986.26
2024	\$223.43M	227,229	\$983.28
2025	\$278.60M	283,573	\$982.46

From 2022 to 2025, revenue increased substantially while AOV remained close to \$980–\$990. This indicates that the long-term revenue expansion was primarily associated with increasing order volume rather than a significant increase in average order value.

3. Product and Category Analysis

3.1 Category Revenue Concentration

Electronics was the dominant category, generating approximately \$732.78 million, equivalent to 74.5% of total revenue.

- Furniture: 11.1%
- Fashion: 6.9%
- Home Goods: 3.4%
- Auto: 2.2%
- Toys: 1.3%
- Books: 0.6%

The top three categories — Electronics, Furniture and Fashion — accounted for approximately 92.6% of total revenue. This demonstrates significant category-level revenue concentration.

3.2 Category Profitability

Category	Revenue Share	Gross Margin	Gross Profit Share
Electronics	74.5%	18.2%	53.0%
Furniture	11.1%	40.7%	17.6%
Fashion	6.9%	62.5%	16.8%
Home Goods	3.4%	49.2%	6.6%
Toys	1.3%	55.3%	2.8%
Auto	2.2%	25.7%	2.2%
Books	0.6%	39.9%	0.9%

Electronics generated the majority of revenue and gross profit, but its gross margin was substantially lower than Fashion, Home Goods and Toys. Fashion provides an important contrast: it contributed only approximately 6.9% of revenue but 16.8% of gross profit, with a gross margin of approximately 62.5%. This indicates that revenue contribution alone does not provide a complete picture of category performance.

3.3 Product Concentration

Product-level revenue was more concentrated than customer-level revenue. The top 1% of products generated approximately 9.6% of revenue, the top 5% generated 40.4%, the top 10% generated 66.7%, and the top 20% generated 80.9%.

The top 10 individual products were all Electronics products. However, each individual product represented only around 0.5% of total revenue. This suggests that concentration is primarily driven by category-level strength and a broad group of high-performing products rather than dependence on one single product.

4. Customer Analysis

4.1 Customer Segments

Consumer customers represented approximately 70.1% of customers and generated approximately 63.7% of revenue. VIP customers represented approximately 10.0% of customers but contributed approximately 18.1% of revenue. Corporate customers represented approximately 19.9% of customers and contributed approximately 18.1% of revenue.

VIP customers therefore generated substantially higher revenue per customer than the other segments. This suggests that customer segmentation can provide useful commercial information beyond simply measuring customer volume.

4.2 Repeat Customers

Repeat customers represented approximately 84.1% of customers but generated approximately 95.5% of revenue. One-time customers represented approximately 15.9% of customers but contributed only approximately 4.5% of revenue.

This is one of the strongest findings in the analysis. The business is therefore highly dependent on customers who purchase repeatedly, making customer retention and repeat purchasing behaviour important areas for future business analysis.

4.3 Customer Revenue Concentration

The top 1% of customers generated approximately 4.7% of revenue, the top 5% generated 16.8%, the top 10% generated 28.3%, and the top 20% generated 46.2%. The relatively modest contribution from the top 1% indicates that revenue is not dominated by a very small number of individual customers.

4.4 Customer Lifetime

Customers with longer observed purchasing lifetimes generally generated more cumulative revenue and completed more orders. Customers with a lifetime of 3+ years generated average revenue of approximately \$4,640 per customer, compared with approximately \$1,107 among customers with a lifetime of 0–90 days.

However, this relationship should not be interpreted as causal. Older customers have simply had more time to place additional orders. The finding nevertheless highlights the importance of analysing customer retention and purchasing lifecycle.

5. Operations and Payment Analysis

5.1 Seller Performance

The business has approximately 500 sellers. The largest individual sellers each contributed only around 0.2% of total revenue. This indicates that the business does not appear to depend heavily on a single seller based on the available revenue distribution.

5.2 Payment Methods

Credit cards represented approximately 50.7% of total payment value, making them the dominant payment method. Other major payment methods included PayPal, Apple Pay, bank transfer, voucher and debit card. Credit-card transactions also had substantially higher average instalment counts than most other payment methods.

5.3 Delivery Performance

Average actual delivery time was approximately 9.42 days, compared with an average estimated delivery time of approximately 9.43 days. The overall late-delivery rate was approximately 9.86%.

Customer review scores showed a clear association with delivery performance. Orders receiving lower review scores generally showed longer delivery times and higher late-delivery rates, while high review scores were associated with shorter delivery times and lower late-delivery rates. This finding should be treated as an association rather than evidence that delivery time directly causes a particular review score.

6. Key Business Findings

Finding 1 — Revenue growth is volume-driven

Revenue increased substantially from 2022 to 2025 while AOV remained relatively stable. This suggests that increasing order volume has been a more important driver of growth than increasing the value of each individual order.

Finding 2 — Electronics is the revenue engine but not the highest-margin category

Electronics generated approximately 74.5% of revenue but had an 18.2% gross margin. Fashion generated only 6.9% of revenue but achieved a gross margin of approximately 62.5%.

Finding 3 — Repeat customers are the core revenue base

Repeat customers generated approximately 95.5% of revenue while representing approximately 84.1% of customers. Customer retention is therefore closely linked to the business revenue base.

Finding 4 — Product revenue is more concentrated than customer revenue

The top 10% of products generated approximately 66.7% of revenue, compared with approximately 28.3% generated by the top 10% of customers.

Finding 5 — Delivery performance is associated with customer satisfaction

Lower review scores were observed alongside longer delivery times and higher late-delivery rates.

7. Business Recommendations

7.1 Improve Electronics Profitability

Investigate supplier and product-level cost structures, pricing strategies, discount levels, product mix and high-volume, low-margin products. The objective would be to improve margin without damaging the category's strong sales volume.

7.2 Develop High-Margin Category Opportunities

Evaluate whether selected high-margin categories such as Fashion have opportunities for increased product range, additional marketing investment, cross-selling and customer-segment targeting. Revenue growth should be evaluated alongside absolute gross profit rather than margin percentage alone.

7.3 Strengthen Customer Retention

Because repeat customers contribute the majority of revenue, retention should remain a key commercial priority. Potential strategies include personalised recommendations, loyalty programmes, targeted promotions, reactivation campaigns and VIP customer benefits.

7.4 Monitor Product Concentration

Monitor stock availability of high-performing products, product-level profitability, dependency on major categories, product lifecycle and underperforming inventory.

7.5 Investigate Late Deliveries

Approximately 9.9% of reviewed orders were classified as late. Further investigation should focus on seller-level delivery performance, geographic differences, shipping routes, product categories, seasonal periods and estimated delivery accuracy.

8. Technical Methodology

8.1 Data Architecture

The source data consisted of eight relational tables covering Customers, Orders, Order Items, Products, Sellers, Payments, Reviews and Geolocation. The database was developed in MySQL using staging tables, data cleaning, validation, primary keys and performance indexes.

8.2 SQL Analysis

- Clean and transform raw data
- Validate primary and business keys
- Validate relationships and orphan records
- Create analytical fact tables
- Create monthly performance tables
- Calculate revenue, orders, AOV and profitability
- Analyse customers, products, sellers, payments and delivery performance

Separate analytical grains were maintained to avoid double-counting caused by one-to-many relationships.

8.3 Python Analysis

Pandas and Python were used for exploratory data analysis and deeper business analysis. Four analytical notebooks were developed: EDA and Business Performance, Customer Analysis, Product Analysis, and Operations and Payment Analysis.

8.4 Power BI

Power BI was used to create an interactive business dashboard covering Executive Overview, Product Analysis, Customer Analysis, and Operations & Payment. A dedicated date dimension and analytical dimensions were used to support consistent filtering and drill-down analysis.

9. Conclusion

The analysis demonstrates a large-scale e-commerce business with strong revenue volume, a substantial repeat-customer base and significant category-level concentration.

The most important opportunities identified are not simply increasing revenue, but understanding how revenue is generated, how profitable that revenue is, which customers drive repeat purchases, and where operational performance affects customer experience.

The combination of SQL, Python and Power BI provides a complete analytical workflow from relational data preparation through business insight and interactive reporting.

Project capabilities: SQL data modelling → data validation → analytical SQL → Python EDA → business analysis → Power BI → data storytelling → business recommendations.